Assignment 2: Designing and Comparing Model Evaluations

Overview:

In a group of **2–3 people**, you will design an evaluation of a safety-relevant model behavior and investigate whether that behavior can be measured using model activations.

The assignment has two connected parts:

1. **Build and validate a behavioral evaluation** using an LLM judge.
2. **Train an activation probe to predict the judge's labels**, then compare the two methods.

The goal is to understand what your evaluation measures, how well a simple probe reproduces its judgments, and what agreement or disagreement between these methods tells you. You do not need to achieve high probe accuracy or discover a large behavioral effect to receive full credit.

Learning Goals:

- Turn a question about model behavior into a testable hypothesis.
- Design and validate an LLM-based scoring procedure.
- Train a simple activation probe.
- Compare evaluation methods and investigate their disagreements.
- Recognize shortcuts, confounds, and limits to generalization.
- Communicate conclusions with scientific rigor.

Setup:

- **Group size:** 2–3 students.
- **Budget:** Aim to spend no more than $150-200 per group.
- **Target model:** Use an open-weight model whose activations you can access.
- **Judge:** You may use an API-accessible or open-weight model to assess the target model's responses.
- **Probe:** Train a probe, such as logistic regression or MLP, on the target model's activations. Keep the target model's weights fixed.

Record model versions, prompts, generation settings, data splits, etc.

Part 1: Design and Interpret an Evaluation

1. Choose a behavior and define your question

Choose **one** of the following topics:

Topic	Example question
Refusal	Does the model refuse the request, partially comply, or substantively comply?
Sycophancy	Does the model endorse a user's incorrect belief despite evidence to the contrary?
Evaluation awareness	Does the response express recognition or suspicion that the current interaction is an evaluation?
Instruction hierarchy	Does the model follow a lower-priority instruction that conflicts with a higher-priority instruction?

*Note: it may be slightly harder to evaluate eval awareness and instruction hierarchy meaningfully without a more elaborate set up, but we encourage you to try as it can teach you a lot about the hard parts of doing eval. We will grade your work with this in mind.

You may propose another behavior of similar scope.

Define the behavior precisely. For example:

- For **evaluation awareness**, judging expressed awareness does not establish whether the model has unverbalized awareness.
- For **instruction hierarchy**, distinguish the presence of conflicting instructions from whether the response actually violates the hierarchy.
- For **refusal**, distinguish whether the model refuses from whether refusing was appropriate.

2. Build your dataset and judge

Create or select prompts that produce a range of responses relevant to your question.

Write a judge prompt with a clear rubric. The judge should assess the response in context, rather than simply look for particular words. A binary label is sufficient, although you may use a small number of clearly defined categories.

Include:

- **A control** addressing at least one plausible alternative explanation.

- **Examples that challenge simple surface cues.** For instance, “I cannot help with that” followed by substantive compliance should not automatically count as a refusal.
- **A manually labeled sample** to check whether the judge applies your rubric correctly.

Inspect disagreements between your manual labels and the judge. Refine the rubric using development examples, and document important ambiguities.

3. Set aside data for final evaluation

Divide your data into **training, validation, and test sets**.

- Use training and validation data to develop the judge and probe.
- Use validation data to select probe settings and classification thresholds.
- Keep the test set untouched until your judge and probe are finalized.

Split by underlying scenario. Paraphrases, variants, or multiple responses to the same scenario should remain in the same split.

4. Run and interpret the behavioral evaluation

Use your judge to label the target model’s responses. Check that you have enough examples of each label for a meaningful prediction task.

Report the observed behavior on development data, with an appropriate measure of uncertainty. Explain:

- What does your evaluation measure?
- What evidence supports your interpretation?
- What alternative explanations remain?
- Which situations do your conclusions cover?

Freeze your judge prompt and rubric before the final probe comparison.

Part 2: Predict the Judge from Activations

1. Extract activations and train a probe

Use the provided infrastructure to extract an activation vector for each example. Collect activations at the **beginning** and **end** of the generated response at multiple layers.

Train a classifier to predict the judge’s label from this activation. Fit preprocessing steps, such as feature standardization, using training data only. Which positions and layers are best / worst?

2. Compare with simple baselines

Compare your probe with:

- **A majority-class baseline:** Always predict the most common training label.
- **A simple text classifier:** For example, logistic regression using bag-of-words or TF-IDF features from the generated response.
- Any other baseline that you think is appropriate.

The text baseline helps you assess whether simple wording cues already explain much of the predictive performance. State what information each method receives, since the judge, activation probe, and text classifier may have different access to context.

Choose metrics appropriate to your label distribution. Include a confusion matrix and per-class precision and recall; overall accuracy alone may be misleading when one label is much more common.

3. Evaluate on held-out examples

Select and freeze your probe configuration using validation results. Then evaluate the judge-label prediction task on the test set.

Include one modest generalization check, such as a held-out prompt template or topic. This can be built into your test split rather than requiring a separate large experiment.

Report whether performance changes and discuss possible explanations.

4. Investigate agreement and disagreement

Manually review a small sample of test examples, including:

- Examples where the judge and probe disagree.
- Examples where they agree.
- Examples representing different labels.

Where possible, assign your manual label before viewing the judge and probe predictions.

Ask:

- Did the probe fail to reproduce a reasonable judgment?
- Did the judge make an error?
- Is the rubric ambiguous?
- Are either of the methods relying on a superficial cue?

Use the broader sample to assess correctness. Disagreement examples alone are useful for diagnosis but do not estimate overall accuracy.

5. Interpret your findings

Discuss:

- How well does the probe predict the judge on unseen scenarios?

- How does it compare with the simple text baseline?
- Does performance survive your generalization check?
- What do human-reviewed examples reveal about the two methods?
- What can you conclude about the behavior, and what remains uncertain?
- For your choice of the probe, what implicit assumptions about the geometry of concept representation are you making?

Recommended Process:

Week 1: October 23–November 2

- Choose your question and define the behavior.
- Run a small pilot to check that your model produces a useful range of examples.
- Build your judge and manually review a sample.
- Identify a plausible confound and implement a control.
- Create training, validation, and test splits.
- Run the baseline behavioral evaluation on development data.
- Finalize your judge rubric and confirm that activation extraction works.

Week 2: November 2–November 9

- Extract activations and train the probes.
- Train the simple text baseline.
- Select and freeze configurations using validation data.
- Run the held-out evaluation and generalization check.
- Manually inspect agreements and disagreements.
- Reserve the final several days for analysis and writing.

Submission:

Submit:

1. **A concise report** describing your question, dataset, judge, controls, probe, comparisons, and conclusions.
2. **Code and configurations** sufficient to reproduce your main results, including the judge rubric and data-splitting procedure.
3. **Representative examples**, including disagreements and cases that complicate your interpretation.

Clearly distinguish exploratory findings from held-out results. Document any changes made after inspecting test results.

Grading:

We will emphasize:

- Clear definitions and well-motivated evaluation design.

- Appropriate controls and validation of judge labels.
- Correct data splitting and probe training.
- Meaningful baseline comparisons.
- Careful analysis of design choices, agreements, disagreements, and limitations.
- Reproducibility and clear communication.